

Notes: Geelen, Hajda, Morellec & Winegar (JFE, 2024)
Asset life, leverage, and debt maturity matching

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1 Big picture

- **Question.** How do the useful life of assets and the age of capital in place shape leverage dynamics and debt maturity choices?
- **Main mechanism.** Capital has a finite useful life. Firms know that old assets will eventually have to be replaced, and they need financial slack at that replacement date.
- **Financing logic.** Firms lever up when buying new capital, then delever as capital ages so they have enough debt capacity when replacement investment arrives.
- **Maturity logic.** Debt is structured so that repayment progressively frees capacity as the asset approaches replacement. This creates maturity matching between debt contracts and useful asset lives.
- **Debt cycles.** The model predicts repeating leverage cycles: debt rises at investment, then falls as the firm frees up capacity, then rises again when assets are replaced.
- **Debt maturity cycles.** Debt maturity also falls with capital age because long-maturity debt issued at investment becomes shorter as the replacement date approaches.
- **Core empirical predictions.** Within firms, capital age should be negatively related to leverage and debt maturity. Across firms, useful asset life should be positively related to debt cycle length and average debt maturity.
- **Main empirical result.** Using U.S. public firms from 1975 to 2018, the paper finds strong support for each of these predictions.
- **Contribution.** The paper reconciles apparently conflicting evidence about maturity matching: asset useful life matters in cross-section, while capital age matters in time-series within firms.
- **Economic takeaway.** Debt maturity matching is not simply collateral matching. It is a dynamic liquidity-management policy that prepares the firm for predictable replacement investment.

2 Data and measurement

2.1 Sample and core variables

- The empirical sample covers listed U.S. firms from 1975 to 2018.
- Capital age is the average age of the firm's capital stock, calculated from investment and depreciation data.
- Useful life of capital proxies the expected life of newly installed assets.
- Leverage is measured as net debt to EBITDA, net book leverage, and net market leverage.
- Debt maturity is measured as the share of debt maturing after three years, the share maturing after five years, and average debt maturity from Capital IQ.

Interpretation: The paper separates two concepts often conflated in maturity-matching tests: the age of capital already in place and the useful life of newly installed capital. Capital age is a dynamic state variable, while useful life is a stable horizon variable.

2.2 Capital age

- Capital age rises as the firm's asset base gets older and falls after investment replaces old capital.
- The model predicts a within-firm negative relation between capital age and both leverage and debt maturity.

- Empirically, capital age is useful because it varies over time within a firm, making it compatible with firm fixed-effect regressions.

Interpretation: Capital age tells us where the firm is in its investment and debt cycle. A firm with young assets has recently invested and may have high debt. A firm with old assets is closer to replacement and should have lower net debt and shorter remaining maturity.

2.3 Useful life of assets

- Useful life is closer to a stable firm or industry characteristic.
- It captures whether a firm's capital base tends to last a long time before replacement is needed.
- The model predicts that longer useful life leads to longer debt cycles and longer average debt maturity.

Interpretation: Useful life sets the natural horizon of the financing cycle. Firms with long-lived assets can issue longer debt and follow longer cycles; firms with short-lived assets should recycle debt capacity more quickly.

2.4 Debt cycles

- The empirical debt cycle is built around peaks and troughs in net debt to EBITDA.
- The cycle length measures how long it takes for firms to move from debt peaks toward troughs.
- Event-time plots show investment, capital age, and net debt around leverage peaks.

Interpretation: Debt cycles operationalize the model mechanism: investment creates a leverage peak; subsequent deleveraging prepares the firm for the next replacement investment.

2.5 Controls and empirical benchmark

- Leverage regressions control for profitability, size, market-to-book, tangibility, cash flow volatility, R&D, and firm age.
- Debt maturity regressions control for standard debt maturity determinants including size, market-to-book, asset maturity, abnormal earnings, cash flow volatility, R&D, net book leverage, and firm age.
- Firm fixed effects absorb time-invariant firm characteristics; year and industry-year fixed effects absorb macro and industry shocks.

Interpretation: The tests are designed to show that capital age and useful life have explanatory power beyond standard capital-structure and debt-maturity variables.

3 Model and empirical specifications

3.1 Finite asset life and replacement investment

Each unit of capital produces profit π each period for a finite useful life n . At the end of useful life, the capital must be replaced at cost K . The value of a firm that replaces capital every n periods is:

$$\sum_{t=1}^{\infty} \frac{\pi}{(1+r)^t} - \sum_{i=0}^{\infty} \frac{K}{(1+r)^{in}} = \frac{\pi}{r} - \frac{(1+r)^n K}{(1+r)^n - 1}.$$

Interpretation: This expression makes replacement investment explicit. The firm knows that keeping production going requires large, lumpy investment every n periods. This predictable investment need is what gives capital age economic content.

3.2 Cash-flow-based borrowing constraint

The firm faces a borrowing constraint tied to cash flow:

$$D_t \leq \phi\pi.$$

Interpretation: The constraint is cash-flow based rather than collateral based. This matters because the model can generate debt and maturity cycles even when the value of collateral is not the only binding financing force. The firm repays debt because future investment needs debt capacity.

3.3 Dividends and net debt

The dividend constraint can be summarized as:

$$Div_t = \pi \mathbf{1}_{\{firm \text{ produces}\}} - I_t + C_{t-1}(1 + \rho_C) - C_t + D_t - D_{t-1}(1 + \rho_D) - e \max\{D_t - D_{t-1}, 0\}.$$

Net debt is:

$$ND_t = D_t - C_t.$$

Interpretation: Net debt summarizes the joint choice of debt and cash. Deleveraging can occur by repaying debt or accumulating cash, both of which increase future financing capacity.

3.4 Debt cycles and maturity matching

The model implies that as capital ages, net debt falls:

$$ND_{a+1} \leq ND_a.$$

Debt maturity also declines with capital age:

$$M_{a+1} \leq M_a.$$

Interpretation: After investment, debt is high. As the asset ages, the firm optimally reduces net debt and the remaining debt becomes shorter maturity. This creates declining leverage and maturity over the capital-age cycle.

3.5 Within-firm regressions

The leverage regression is:

$$Lev_{i,j,t+1} = \phi CapitalAge_{i,t} + \beta X_{i,t} + \eta_i + \gamma_{t+1} + \kappa_{j,t+1} + \epsilon_{i,j,t+1}.$$

The debt maturity regression is:

$$Mat_{i,j,t+1} = \phi CapitalAge_{i,t} + \beta X_{i,t} + \eta_i + \gamma_{t+1} + \kappa_{j,t+1} + \epsilon_{i,j,t+1}.$$

Interpretation: The coefficient ϕ is the main object. The theory predicts $\phi < 0$ in both leverage and maturity regressions. Firm fixed effects mean the coefficient is identified from how a given firm changes as its capital stock ages.

3.6 Cross-sectional debt-cycle and maturity regressions

The cross-sectional debt-cycle regression is:

$$CycleLength_i = \alpha + \phi UsefulLife_i + \beta X_i + \epsilon_i.$$

The cross-sectional maturity regression is:

$$AverageMaturity_i = \alpha + \phi UsefulLife_i + \beta X_i + \epsilon_i.$$

Interpretation: The theory predicts $\phi > 0$: firms with longer-lived assets should follow longer debt cycles and issue longer-maturity debt. These tests use cross-sectional variation because useful life is relatively stable over time.

4 Identification and empirical design

4.1 Within-firm identification

- Capital age varies over time within a firm.
- Firm fixed effects absorb persistent business-model, industry, and asset-type differences.
- The key question is whether the same firm has lower leverage and shorter debt maturity when its capital stock is older.

Interpretation: This design is well suited for capital age, because capital age is a within-firm state variable.

4.2 Cross-sectional identification

- Useful life of capital is mainly a firm or industry characteristic.
- Because useful life changes little over time, firm fixed effects would remove much of its variation.
- The paper therefore uses cross-sectional tests to connect useful life to debt cycle length and average debt maturity.

Interpretation: The paper's reconciliation of previous maturity-matching studies comes from separating cross-sectional useful-life variation from time-series capital-age variation.

4.3 Mechanism tests

- The model predicts stronger effects when investment is lumpier.
- The model predicts stronger effects when return on investment is lower.
- Smaller firms may also show stronger effects because financing constraints are more binding.
- The paper tests these predictions using interaction specifications and tercile splits.

Interpretation: These tests are important because they check not only whether capital age matters, but whether it matters in the circumstances the model says it should.

4.4 Robustness logic

- The paper tests alternative samples, alternative depreciation-rate definitions, and alternative capital-age measures.
- It also studies alternative debt maturity and debt-cycle measures in the appendix.
- The core signs remain stable across the main robustness exercises.

Interpretation: Robustness reduces concern that the results are artifacts of a particular accounting definition of capital age.

4.5 Main limitations

- The tests are not randomized experiments.
- Capital age may be correlated with other unobserved time-varying firm states.
- Useful life is partly inferred from accounting data and depreciation rates.
- Debt maturity may reflect many contractual motives besides maturity matching, such as refinancing risk, information asymmetry, credit ratings, and market timing.

Interpretation: The paper's strength is that a coherent theory generates multiple linked predictions, and all of them line up with the empirical evidence.

5 Tables and figures with full interpretation

5.1 Figure 1: Descriptive relations linking capital age, useful life, leverage, debt maturity, and cycle length

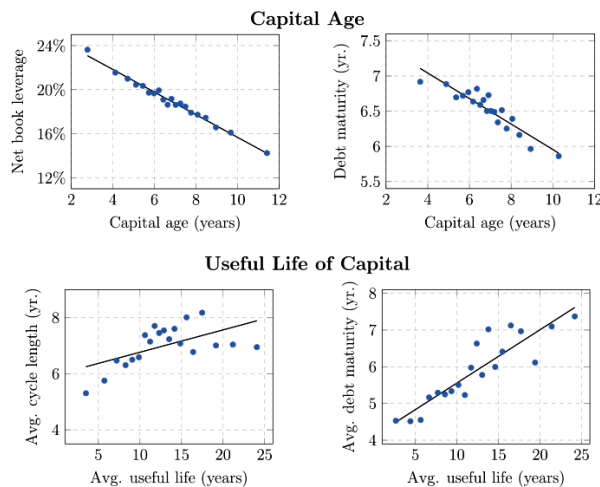
Read:

- The top-left panel shows a negative relation between capital age and net book leverage.
- The top-right panel shows a negative relation between capital age and debt maturity.
- The bottom-left panel shows a positive relation between useful life of capital and average debt cycle length.
- The bottom-right panel shows a positive relation between useful life and average debt maturity.
- The top panels control for firm fixed effects; each dot represents one-twentieth of the sample firms.

Interpretation: Figure 1 is the paper in miniature. Capital age is the within-firm state variable that predicts where the firm is in its financing cycle. Useful life is the cross-sectional asset characteristic that predicts how long the cycle should be and how long debt maturity should be. The top and bottom rows therefore represent two different empirical sources of variation.

Economic message: The descriptive evidence already supports the maturity-matching mechanism: firms delever and shorten debt maturity as capital ages, while firms with longer-lived capital have longer debt cycles and longer debt maturity.

What not to over-read: This figure is descriptive. It motivates the tests but does not by itself establish causality or rule out omitted time-varying firm characteristics.



· firm fixed effects. Each dot corresponds to 1/20th of the sample firms. The sample peric

Figure 1: descriptive relations. Directly cropped from the paper.

5.2 Figure 2 and Figure 3: The model mechanism and debt cycles

Read:

- Figure 2 shows the timing of profits and replacement investment when capital has finite useful life.
- Figure 3 shows a firm that invests in lumps, uses debt at investment dates, and gradually lowers net debt as capital ages.
- The firm produces profits each period but must periodically pay a large replacement cost K .
- Net debt jumps at investment and then declines until the next replacement date.

Interpretation: The figures clarify why finite asset life changes financing decisions. The firm cannot treat debt capacity as something to use only today; it must preserve enough capacity for the next replacement investment. The resulting optimal policy resembles a planned deleveraging schedule rather than passive adjustment toward a target leverage ratio.

Economic message: Debt cycles arise because investment is lumpy and predictable. Maturity matching arises because liabilities are structured to free debt capacity when the next asset replacement is due.

What not to over-read: The figures are stylized. Real firms hold many asset vintages, face shocks, and can refinance. The point is the mechanism, not a literal one-asset cycle.

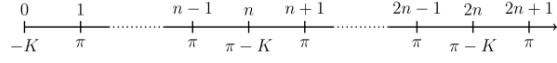


Figure 2: cash-flow and replacement timeline.

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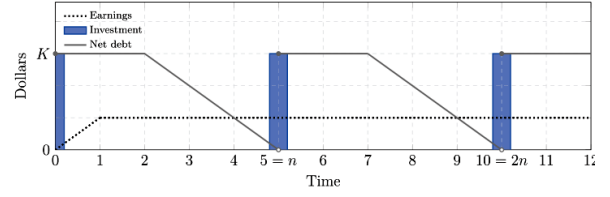


Figure 3: earnings, investment, and net debt dynamics.

5.3 Figure 4 and Figure 5: Optimal debt maturity and the maturity cycle

Read:

- Figure 4 shows staggered debt issues that mature at different times as the asset approaches replacement.
- The firm does not simply issue one-period debt; it uses a maturity structure that coordinates repayment with future investment needs.
- Figure 5 shows average debt maturity falling through the asset life and resetting when new capital is purchased.
- Together, the figures show both the level of leverage and the maturity of debt moving with capital age.

Interpretation: The key insight is that debt maturity is not chosen only to reduce rollover risk or adverse selection. In the model, maturity is a commitment to return debt capacity to the firm before replacement investment arrives. As the debt ages and amortizes, average maturity declines. When the firm invests again, it issues new longer-maturity debt and the cycle resets.

Economic message: Maturity matching is a dynamic liquidity policy: the firm matches liability maturity to asset life because it wants cash-flow capacity at the replacement date.

What not to over-read: These figures do not imply every bond should mature exactly when the asset expires. They show an optimal maturity structure in a stylized environment with issuance costs and predictable replacement.

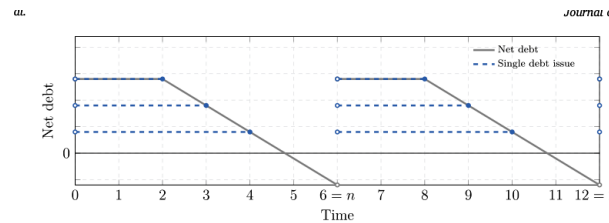


Figure 4: firm financing and optimal debt maturity.

Figure 4: firm financing and optimal debt maturity.

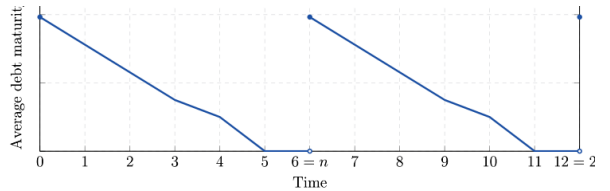


Fig. 5. Average debt maturity.

Figure 5: average debt maturity through the cycle.

Figure 5: average debt maturity through the cycle.

5.4 Figure 6 and Figure 7: Investment frequency and the shape of leverage cycles

Read:

- Figure 6 compares debt cycles when investment occurs every n periods versus every $n/2$ periods.

- More frequent investment creates shorter and less pronounced cycles.
- Figure 7 shows that leverage dynamics are smoother when investments are more frequent and smaller.
- Less frequent and lumpier investment generates stronger debt reduction between investment dates.

Interpretation: These figures explain why the effect of capital age should be stronger for firms with lumpier investment. If replacement is a large rare event, the firm must prepare by freeing substantial debt capacity. If replacement is frequent and divisible, the need for large pre-funding capacity is weaker, so the capital-age pattern is less pronounced.

Economic message: The model predicts heterogeneity: capital age should matter most when investment is lumpy and replacement needs are large.

What not to over-read: The figures abstract from unplanned investment and shocks. In the data, the mechanism should be visible statistically rather than as perfectly regular cycles.

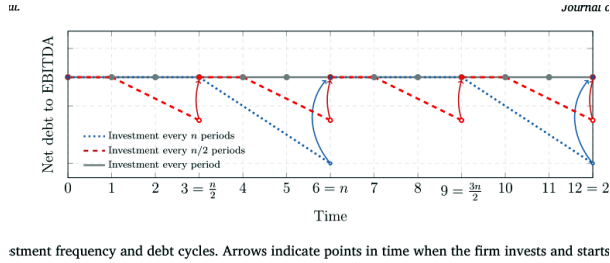
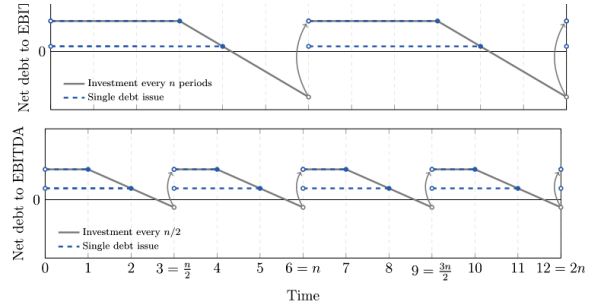


Figure 6: investment frequency and debt cycles.



for different investment frequencies. The higher the frequency of investment, the less pronounced

is fixed investment costs. Also on the actual path of the TED cycle.

Figure 7: leverage dynamics under different investment frequencies.

5.5 Table 1 and Figure 8: Summary statistics, correlations, and an event-time debt cycle

Read:

- Table 1 reports average capital age of 6.804 years and average useful life of 13.048 years.
- The average net debt to EBITDA ratio is 2.236, and average debt maturity is 6.514 years.
- Within-firm correlations show capital age negatively correlated with net book leverage, net market leverage, and debt maturity measures.
- Figure 8 plots investment, capital age, and net debt to EBITDA around leverage peaks.
- Investment rises before the leverage peak; after the peak, capital age rises and net debt to EBITDA falls.

Interpretation: Table 1 shows that the variables have substantial dispersion and that raw within-firm correlations already line up with the model. Figure 8 adds event-time discipline: leverage peaks appear shortly after investment, and the subsequent fall in leverage occurs while capital age rises. This is exactly the debt-cycle pattern implied by replacement investment.

Economic message: The summary statistics and event study make the theory empirically plausible before the formal regressions.

What not to over-read: Pairwise correlations and event-time averages do not control for all covariates. They are suggestive diagnostics, not the final evidence.

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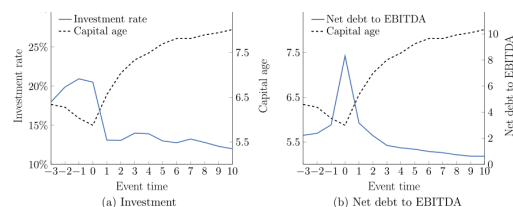
Table 1

Summary statistics: capital age and financing. The table contains the summary statistics of capital age, the useful life of assets, and the financing variables. These include three measures of net leverage: net debt to EBITDA, net book leverage, net market leverage and three measures of debt maturity: the ratios of debt maturing in more than 3 or 5 years to total debt as well as the debt maturity from Capital IQ. Panel A contains the summary statistics and Panel B contains the within-firm pairwise correlations between the respective variables. All variables are defined in Table C.1.

Panel A: Summary statistics								
	Capital age	Useful life	ND/EBITDA	Net book leverage	Net mkt. leverage	% debt mat.> 3y	% debt mat.> 5y	Debt mat. (yr.)
Mean	6.804	13.048	2.236	0.189	0.223	0.520	0.328	6.514
Standard deviation	3.215	5.709	4.205	0.226	0.264	0.328	0.302	4.861
Q1	4.408	9.000	0.312	0.050	0.041	0.228	0.005	3.337
Median	6.367	13.000	1.441	0.200	0.199	0.584	0.292	5.288
Q3	8.700	17.000	3.033	0.340	0.397	0.795	0.569	8.109
N	68833	66380	68833	68833	68833	68833	68833	16905

Panel B: Within-firm pairwise correlations								
	Capital age	Useful life	ND/EBITDA	Net book leverage	Net mkt. leverage	% debt mat.> 3y	% debt mat.> 5y	Debt mat. (yr.)
Capital age	1							
Useful life	0.243	1						
ND/EBITDA	-0.041	0.001	1					
Net book lev.	-0.144	-0.074	0.507	1				
Net mkt. lev.	-0.121	-0.058	0.518	0.843	1			
% debt mat.> 3y	-0.098	-0.005	0.032	0.139	0.078	1		
% debt mat.> 5y	-0.120	0.013	0.016	0.085	0.049	0.640	1	
Debt mat. (yr.)	-0.065	0.008	0.005	0.031	0.004	0.202	0.250	1

Table 1: summary statistics and within-firm correlations.



to trough. Dynamics of capital age, investment, and net debt to EBITDA around a net debt to EBITDA peak. Event We include debt cycles with at least 3 years from peak to trough, defined as the year in which net debt to EBITDA is defined in Table C.1.

the academic literature. We test the mediations of capital age from Capital IQ to EBITDA using its time with net

Figure 8: empirical debt cycle from peak to trough.

5.6 Table 2: Capital age and leverage within firms

Read:

- Capital age is negative and highly significant across all leverage measures.
- For net debt to EBITDA, coefficients range from -0.355 to -0.425 depending on controls and fixed effects.
- For net book leverage, coefficients are around -0.032 to -0.041.
- For net market leverage, coefficients are around -0.031 to -0.042.
- The estimates remain significant after adding standard leverage controls and industry-year fixed effects.

Interpretation: Table 2 is the first formal within-firm test of the model. Since all variables are standardized, the coefficients can be interpreted as the effect of a one-standard-deviation increase in capital age. The negative signs mean that as the same firm's capital stock gets older, leverage falls. The result is strongest for net debt to EBITDA, the measure closest to cash-flow-based borrowing constraints.

Economic message: Older capital is associated with lower debt capacity usage. Firms appear to delever as replacement investment approaches.

What not to over-read: The table does not prove that capital age alone causes deleveraging. But fixed effects and controls reduce the concern that the result is driven purely by stable firm characteristics.

5.7 Table 3: Capital age and debt maturity within firms

Read:

- Capital age is negative and significant for the share of debt maturing after three years.
- Capital age is negative and significant for the share of debt maturing after five years.
- Capital age is negative and significant for debt maturity from Capital IQ.

Table 2

Capital age and leverage – within-firm regressions. This table presents estimates from regressions of net debt to EBITDA and net leverage ratios on lagged capital age. The dependent variable is *Net debt to EBITDA* in columns 1 to 3; *Net book leverage* in columns 4 to 6 and *Net market leverage* in columns 7 to 9. Each explanatory variable is standardized by its full-sample standard deviation. The models in columns 3, 6 and 9 include industry-year fixed effects created using Hoberg-Phillips fixed industry classification with 100 industries. All variables are defined in Table C.1. *t*-statistics are reported in parentheses. Standard errors are clustered at the firm level. We use * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ to indicate *p*-values.

	ND/EBITDA			Net book leverage			Net market leverage		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Capital age	-0.425*** (-11.46)	-0.355*** (-8.48)	-0.403*** (-7.55)	-0.041*** (-17.26)	-0.032*** (-11.57)	-0.032*** (-9.54)	-0.042*** (-9.91)	-0.031*** (-9.09)	-0.033*** (-9.05)
Profitability		-0.817*** (-22.90)	-0.745*** (-17.11)		-0.032*** (-18.20)	-0.027*** (-12.38)		-0.048*** (-24.11)	-0.045*** (-18.25)
Size		0.484*** (4.50)	0.516*** (3.38)		0.050*** (8.35)	0.062*** (7.15)		0.085*** (10.82)	0.100*** (10.40)
Market-to-book		-0.049 (-1.57)	-0.044 (-1.21)		-0.009*** (-4.39)	-0.012*** (-5.01)		-0.026*** (-12.74)	-0.023*** (-10.40)
Tangibility		0.371*** (5.24)	0.392*** (4.15)		0.039*** (7.90)	0.040*** (7.14)		0.048*** (8.95)	0.045*** (7.26)
Cash flow volatility		-0.042 (-1.28)	-0.020 (-0.55)		-0.003*** (-2.13)	-0.001 (-0.55)		-0.003*** (-1.97)	-0.000 (-0.11)
R&D		-0.116*** (-2.14)	-0.010 (-0.17)		-0.007 (-1.79)	-0.001 (-0.32)		-0.008*** (-2.11)	-0.001 (-0.24)
Firm age		0.140 (0.29)	0.772 (1.57)		-0.024 (-0.62)	0.033 (0.96)		-0.020 (-0.53)	0.020 (0.54)
Year FE	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-Year FE	No	No	Yes	No	No	Yes	No	No	Yes
Observations	56707	48261	32499	56707	48261	32499	56707	48261	32499
Adj. within R^2	0.0063	0.0450	0.0411	0.0303	0.0847	0.0812	0.0206	0.1179	0.1163

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Table 3

Capital age and debt maturity – within-firm regressions. The table presents estimates from regressions of debt maturity on lagged capital age. The dependent variable is % of debt maturing in > 3 years in columns 1 to 3; % of debt maturing in > 5 years in columns 4 to 6; and Debt maturity (yr.) in columns 7 to 9. Each explanatory variable is standardized by its full-sample standard deviation. Models in columns 3, 6 and 9 include industry-year fixed effects created using Hoberg-Phillips fixed industry classification with 100 industries. All variables are defined in Table C.1. *t*-statistics are reported in parentheses. Standard errors are clustered at the firm level. We use * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ to indicate *p*-values.

	% debt maturing > 3y			% debt maturing > 5y			Debt maturity (yr.)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Capital age	-0.039*** (-13.42)	-0.030*** (-8.96)	-0.030*** (-6.57)	-0.031*** (-10.21)	-0.025*** (-7.00)	-0.021*** (-4.50)	-0.390*** (-2.97)	-0.413*** (-2.94)	-0.510*** (-3.35)
Size		0.105*** (3.94)	0.189*** (5.31)		0.023 (0.91)	0.070** (2.22)		1.919** (1.97)	1.694 (1.44)
Size squared		-0.051*** (-2.07)	-0.128*** (-3.90)		0.023 (0.96)	-0.009 (-0.30)		-1.230 (-1.20)	-0.944 (-0.79)
Market-to-book		0.007*** (2.39)	0.007*** (1.96)		0.004 (1.57)	0.001 (0.41)		0.077 (0.78)	0.068 (0.60)
Asset maturity		0.006 (1.40)	0.003 (0.66)		0.007*** (1.65)	0.006 (0.97)		0.212 (1.53)	0.181 (1.12)
Abnormal earnings		0.002*** (2.13)	0.001 (0.84)		0.003*** (2.70)	0.003*** (2.28)		0.046** (2.05)	0.064** (2.04)
Cash flow volatility		-0.001 (-0.46)	0.002 (0.62)		-0.002 (-0.07)	0.002 (0.76)		0.005 (0.07)	0.002 (0.03)
R&D		-0.007 (-1.33)	-0.010 (-1.52)		-0.007 (-1.31)	-0.008 (-1.08)		0.162 (0.92)	0.034 (0.17)
Net book leverage		0.032*** (9.43)	0.040*** (9.02)		0.015*** (4.45)	0.016*** (3.69)		0.084 (0.88)	0.150 (1.41)
Firm age		-0.062 (-1.28)	-0.031 (-0.61)		-0.094* (-1.71)	-0.048 (-0.76)		2.461 (1.30)	2.830 (1.43)
Year FE	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-Year FE	No	No	Yes	No	No	Yes	No	No	Yes

Table 3

Capital age and debt maturity – within-firm regressions. The table presents estimates from regressions of debt maturity on lagged capital age. The dependent variable is % of debt maturing in > 3 years in columns 1 to 3; % of debt maturing in > 5 years in columns 4 to 6; and Debt maturity (yr.) in columns 7 to 9. Each explanatory variable is standardized by its full-sample standard deviation. Models in columns 3, 6 and 9 include industry-year fixed effects created using Hoberg-Phillips fixed industry classification with 100 industries. All variables are defined in Table C.1. *t*-statistics are reported in parentheses. Standard errors are clustered at the firm level. We use * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ to indicate *p*-values.

	% debt maturing > 3y			% debt maturing > 5y			Debt maturity (yr.)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Capital age	-0.039*** (-13.42)	-0.030*** (-8.96)	-0.030*** (-6.57)	-0.031*** (-10.21)	-0.025*** (-7.00)	-0.021*** (-4.50)	-0.390*** (-2.97)	-0.413*** (-2.94)	-0.510*** (-3.35)
Size		0.105*** (3.94)	0.189*** (5.31)		0.023 (0.91)	0.070** (2.22)		1.919** (1.97)	1.694 (1.44)
Size squared		-0.051*** (-2.07)	-0.128*** (-3.90)		0.023 (0.96)	-0.009 (-0.30)		-1.230 (-1.20)	-0.944 (-0.79)
Market-to-book		0.007*** (2.39)	0.007*** (1.96)		0.004 (1.57)	0.001 (0.41)		0.077 (0.78)	0.068 (0.60)
Asset maturity		0.006 (1.40)	0.003 (0.66)		0.007*** (1.65)	0.006 (0.97)		0.212 (1.53)	0.181 (1.12)
Abnormal earnings		0.002*** (2.13)	0.001 (0.84)		0.003*** (2.70)	0.003*** (2.28)		0.046** (2.05)	0.064** (2.04)
Cash flow volatility		-0.001 (-0.46)	0.002 (0.62)		-0.002 (-0.74)	0.002 (0.76)		0.005 (0.07)	0.002 (0.03)
R&D		-0.007 (-1.33)	-0.010 (-1.52)		-0.007 (-1.31)	-0.008 (-1.08)		0.162 (0.92)	0.034 (0.17)
Net book leverage		0.032*** (9.43)	0.040*** (9.02)		0.015*** (4.45)	0.016*** (3.69)		0.084 (0.88)	0.150 (1.41)
Firm age		-0.062 (-1.28)	-0.031 (-0.61)		-0.094* (-1.71)	-0.048 (-0.76)		2.461 (1.30)	2.830 (1.43)
Year FE	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-Year FE	No	No	Yes	No	No	Yes	No	No	Yes

- A one-standard-deviation increase in capital age is associated with roughly 0.51 year lower debt maturity in the most saturated specification.
- Asset maturity is not the key time-series determinant once firm fixed effects are included.

Interpretation: Table 3 is the maturity counterpart to Table 2. It shows that the same firm shortens its debt maturity as its capital ages. This is exactly the model's prediction: debt issued at investment becomes shorter and is repaid as the asset approaches replacement. The finding also clarifies why prior fixed-effect maturity-matching tests often failed to find an asset-maturity effect: asset useful life is more cross-sectional, while capital age is the relevant time-series object.

Economic message: Debt maturity is part of the same cycle as leverage. Maturity falls with capital age because debt contracts are designed to free capacity before replacement investment.

What not to over-read: The result does not imply traditional asset maturity is irrelevant. It implies that useful life matters mainly across firms, whereas capital age matters within firms over time.

5.8 Table 4: Mechanism tests

Read:

- Table 4 interacts capital age with terciles of investment skewness, investment kurtosis, return on investment, and firm size.
- The capital-age effect is stronger when investment is more lumpy.
- The capital-age effect is weaker when return on investment is high.
- The table also examines whether smaller firms show stronger financing responses to capital age.
- The strongest evidence is for lumpiness and return on investment, which are the mechanisms most directly tied to the model.

Interpretation: Table 4 tests the mechanism rather than only the baseline correlation. The model says capital age matters because replacement investment is large and because firms must reserve debt capacity for it. Therefore, the effect should be stronger when investment is lumpier and when the cost of investment is large relative to the profits it produces. The interaction patterns support this logic.

Economic message: The capital-age result is not just a generic aging-assets correlation. It becomes stronger in precisely the environments where replacement financing pressure should be more severe.

What not to over-read: Some interactions are noisy, and not every coefficient is significant. The table should be read as mechanism-consistency evidence, not as a complete decomposition of all financing frictions.

N. J. Zingales, E. Mooradian et al.

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Table 4

Exploring the mechanism. This table presents estimates from regressions of net leverage variables and debt maturity on lagged capital age interacted with indicators for firms split into terciles by the proxies of investment lumpiness (the firm-level investment skewness, Panel A; and firm-level investment kurtosis, Panel B), profitability proxied by return on investment (EBITDA divided by book assets, Panel C), and firm size (book assets, Panel D). *Middle* and *High* indicate the middle and highest terciles of each splitting variable. The dependent variables are *Net debt to EBITDA*, *Net Book Leverage*, % of debt maturing in > 3 years, and % of debt maturing in > 5 years. Each explanatory variable is standardized by its full-sample standard deviation. Specifications 2, 4, 6 and 8 control for all independent variables from Tables 2 for net leverage and 3 for debt maturity. *t*-statistics are reported in parentheses. Standard errors are clustered at the firm level. We use * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ to indicate *p*-values.

	ND/EBITDA		Net book leverage		% debt mat. > 3y		% debt mat. > 5y	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Investment lumpiness – Skewness								
Capital age	-0.304*** (-4.17)	-0.203*** (-2.62)	-0.035*** (-7.47)	-0.024*** (-4.86)	-0.033*** (-4.81)	-0.020*** (-2.67)	-0.022*** (-2.97)	-0.011 (-1.43)
Capital age × Middle	-0.163* (-1.74)	-0.137 (-1.46)	-0.006 (-1.04)	-0.005 (-0.88)	-0.009 (-0.96)	-0.007 (-0.79)	-0.007 (-0.78)	-0.008 (-0.85)
Capital age × High	-0.410*** (-3.94)	-0.401*** (-3.82)	-0.016** (-2.48)	-0.017*** (-2.63)	-0.019*** (-2.06)	-0.019*** (-1.97)	-0.015 (-1.61)	-0.019* (-1.93)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	33996	32056	33996	32056	33996	31075	33996	31075
Adj. within R ²	0.0104	0.0437	0.0330	0.0827	0.0080	0.0205	0.0047	0.0092
Panel B: Investment lumpiness – Kurtosis								
Capital age	-0.288*** (-3.62)	-0.231*** (-2.74)	-0.033*** (-6.89)	-0.023*** (-4.70)	-0.041*** (-5.56)	-0.030*** (-3.98)	-0.024*** (-3.32)	-0.015* (-1.88)
Capital age × Middle	-0.174* (-1.85)	-0.076 (-0.80)	-0.010 (-1.64)	-0.006 (-0.89)	0.004 (0.42)	0.008 (0.90)	-0.006 (-0.60)	-0.006 (-0.55)
Capital age × High	-0.434*** (-4.05)	-0.369*** (-3.40)	-0.018*** (-2.80)	-0.018*** (-2.73)	-0.009 (-0.99)	-0.007 (-0.72)	-0.010 (-1.12)	-0.012 (-1.23)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	33996	32056	33996	32056	33996	31075	33996	31075
Adj. within R ²	0.0106	0.0435	0.0332	0.0827	0.0080	0.0205	0.0047	0.0091
Panel C: Return on investment								
Capital age	-0.711*** (-9.87)	-0.642*** (-8.15)	-0.047*** (-13.09)	-0.037*** (-9.54)	-0.041*** (-8.51)	-0.031*** (-5.84)	-0.034*** (-6.62)	-0.028*** (-5.09)
Capital age × Middle	0.295*** (4.95)	0.322*** (5.17)	0.002 (0.71)	0.002 (0.95)	-0.003 (-0.69)	0.003 (0.68)	0.003 (0.62)	0.008 (1.58)
Capital age × High	0.413*** (5.92)	0.451*** (6.10)	0.014*** (3.73)	0.015*** (3.85)	0.001 (0.23)	0.005 (0.73)	0.010* (1.80)	0.015** (2.58)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	34713	32499	34713	32499	34713	31502	34713	31502
Adj. within R ²	0.1125	0.1213	0.0627	0.0967	0.0095	0.0217	0.0056	0.0100
Panel D: Firm size								
Capital age	-0.601*** (-7.26)	-0.592*** (-6.83)	-0.050*** (-10.06)	-0.044*** (-8.51)	-0.044*** (-6.62)	-0.038*** (-5.20)	-0.031*** (-5.16)	-0.030*** (-4.49)
Capital age × Middle	0.223** (2.56)	0.282*** (3.18)	0.016*** (3.04)	0.017*** (3.09)	0.010 (1.27)	0.012 (1.36)	0.007 (1.01)	0.008 (0.96)
Capital age × High	0.213** (2.28)	0.324*** (3.41)	0.019*** (3.29)	0.023*** (3.79)	0.020*** (2.39)	0.023*** (2.61)	0.022*** (2.63)	0.027*** (3.06)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	34713	32499	34713	32499	34713	31502	34713	31502
Adj. within R ²	0.0126	0.0434	0.0496	0.0914	0.0180	0.0259	0.0153	0.0165

Table 4: mechanism tests using investment lumpiness, ROI, and size.

5.9 Tables 5 and 6: Robustness to samples, depreciation definitions, and capital-age measures

Read:

- Table 5 shows that the baseline results remain when R&D-intensive industries are included.

- Table 8 shows that useful life is positively related to the share of debt maturing after three years, the share maturing after five years, and average debt maturity.
- Average capital age also matters in cross-section, but useful life is the key cross-sectional maturity-matching variable.
- The cross-sectional tests complement the within-firm regressions by using stable differences in asset life across firms.

Interpretation: Tables 7 and 8 confirm the long-horizon predictions of the model. Firms with longer-lived assets do not merely have older capital; they also have longer debt cycles and longer debt maturity. This is the classic maturity-matching implication, but expressed in a way that separates asset useful life from the time-varying capital-age state.

Economic message: Useful life sets the horizon of debt cycles and maturity matching. Longer asset life leads to longer financing cycles and longer debt maturity.

What not to over-read: Cross-sectional tests are more exposed to omitted firm-type differences than within-firm tests. The value of these tables is that they match the specific cross-sectional predictions generated by the model.

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Table 7

Asset life and debt cycles – cross-sectional regressions. The dependent variable is *Maximum debt cycle length* in columns 1 and 2, and *Average debt cycle length* in columns 3 and 4. Firms with no leverage spike have a cycle length that is undefined and are dropped from the sample. We require a minimum of three years between subsequent spikes. In specifications 2 and 4 we control for all independent variables from Table 2. *t*-statistics are reported in parentheses. Standard errors are clustered at the industry level using Hoberg-Phillips fixed industry classification with 100 industries. We use * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ to indicate p -values.

	Max debt cycle		Avg. debt cycle	
	(1)	(2)	(3)	(4)
Panel A: Useful life				
Useful life	0.187*** (6.13)	0.104*** (4.06)	0.116*** (5.05)	0.078*** (3.96)
Controls	No	Yes	No	Yes
Observations	2401	2390	2401	2390
Adj. R^2	0.027	0.244	0.017	0.168
Panel B: Average capital age				
Capital age	0.680*** (11.47)	0.176*** (4.50)	0.427*** (11.51)	0.119*** (3.82)
Controls	No	Yes	No	Yes
Observations	2402	2391	2402	2391
Adj. R^2	0.089	0.244	0.057	0.167
Panel C: Asset maturity				
Asset maturity	0.090** (2.63)	0.101*** (3.09)	0.051** (2.17)	0.061** (2.61)
Controls	No	Yes	No	Yes
Observations	2362	2351	2362	2351
Adj. R^2	0.010	0.242	0.005	0.165

industries. The main parameter of interest is the parameter ϕ , which we expect to be positive.

Table 7 presents the resulting estimates for the maximum debt cycle

Table 7: asset useful life and debt cycle length.

Table 8

Asset life and debt maturity – cross-sectional regressions. The dependent variable is the average of each firm's % of debt maturing in > 3 years in columns 1 to 2; % of debt maturing in > 5 years in columns 3 to 4; and *Debt maturity (yr.)* in columns 5 to 6. In specifications 2, 4 and 6 we control for all independent variables from Table 3, except for asset maturity. *t*-statistics are reported in parentheses. Standard errors are clustered at the industry level using the Hoberg-Phillips fixed industry classification with 100 industries. We use * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ to indicate p -values.

	% debt mat. > 3y		% debt mat. > 5y		Debt maturity (yr.)	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Useful life						
Useful life	0.015*** (11.07)	0.010*** (8.66)	0.014*** (8.65)	0.009*** (8.56)	0.150*** (5.82)	0.105*** (5.08)
Controls	No	Yes	No	Yes	No	Yes
Observations	4360	4240	4360	4240	2436	2408
Adj. R^2	0.088	0.421	0.097	0.355	0.044	0.205
Panel B: Average capital age						
Capital age	0.021*** (6.72)	0.008*** (2.85)	0.022*** (9.40)	0.008*** (3.64)	0.236*** (5.80)	-0.005 (-0.12)
Controls	No	Yes	No	Yes	No	Yes
Observations	4376	4247	4376	4247	2441	2411
Adj. R^2	0.041	0.388	0.060	0.321	0.024	0.186
Panel C: Asset maturity						
Asset maturity	0.012*** (8.54)	0.009*** (10.26)	0.010*** (5.62)	0.008*** (10.92)	0.115*** (4.43)	0.105*** (7.24)
Controls	No	Yes	No	Yes	No	Yes
Observations	4301	4175	4301	4175	2397	2368
Adj. R^2	0.086	0.439	0.080	0.359	0.042	0.220

(Panels B and C). As previously noted, the results suggest that asset maturity is a significant determinant of debt maturity in the cross-section while capital age is a key driver of debt maturity in the time-series (Table 3), consistent with our model.

As a robustness test, we show in Table IA.5 in the Internet Appendix that our cross-sectional results for debt cycles are robust to defining

Table 8: asset useful life and debt maturity.

5.11 Appendix Tables C.1 and C.2

Read:

- Table C.1 defines the main variables used throughout the paper.
- Table C.2 reports additional summary statistics for variables not included in Table 1.
- These appendix tables are useful for interpreting capital age, useful life, debt maturity, and alternative proxies.

Interpretation: The appendix tables are not the main empirical tests, but they are essential for replicability. The definitions clarify why the paper distinguishes capital age from useful life and why different debt maturity measures are used.

Economic message: Transparent variable construction is central because the paper’s contribution depends on measuring capital age and useful life distinctly.

and security- and firm-level data, we use the most recent filing dates and remove any observations with the same ID/date, description, maturity, and interest rate. We further remove all securities with missing *gvkey* and drop entries for credit lines that reflect the drawdown limit only, as opposed to actual utilisation. We drop all observations with missing or negative maturity values. We then compute the firm-level maturity as the weighted average of individual-security maturities weighted by their notional amounts. As the final data filter, we drop observations for which the total debt in Capital IQ is greater than Compustat by more than 10%, as in Colla et al. (2013) and Choi et al. (2018).

C.2. Definitions of variables

The variables used in the paper are defined in Table C.1.

Table C.1
Definitions of variables. The table contains the definitions of all variables used throughout the paper (in order of appearance).

Variable	Definition
Capital age	See Subsection 3.1
Useful life	See Subsection 3.1
Net debt to EBITDA	Ratio of total debt ($d1tt+d1c$) less cash (cba) over EBITDA ($ebitda$); set to missing when EBITDA is negative
Net market leverage	Ratio of total debt ($d1tt+d1c$) less cash (cba) over total debt plus market value of equity ($prcc_f+csho$)
Net book leverage	Ratio of total debt ($d1tt+d1c$) less cash (cba) over total assets (act)
% debt maturing > 3y	Ratio of long-term debt ($d1tt$) minus debt maturing in 2- and 3-years ($dd2+dd3$) over total debt ($d1c+d1tt$)
% debt maturing > 5y	Ratio of long-term debt ($d1tt$) minus debt maturing in 2-, 3-, 4-, and 5-years ($dd2+dd3+dd4+dd5$) over total debt ($d1c+d1tt$)
Debt maturity (yr.)	Average maturity of outstanding bonds and loans from Capital IQ, weighted by their notional amounts
Investment	Capital expenditures ($cappx$) over lagged installed capital (i_ppcap)
Profitability	Operating income ($oidbp$) over total assets (act)
Size	Natural log of real sales ($\log(sale/defl)$), where $defl$ is the CPI deflator
Tangibility	Ratio of property, plant and equipment ($ppent$) to total assets (act)
Market-to-book	Ratio of the sum of market value of equity ($prcc_f+csho$) and book value of debt ($act-cba$) to total assets (act)
Cash flow volatility	Moving 3-year standard deviation of profitability
R&D	Ratio of R&D expenditure (xrd) to sales ($sale$), missing values replaced with zero
Firm age	Time since listing (defined as the first appearance of each firm in CRSP) in years
Asset maturity	Gross property, plant and equipment over depreciation and amortization ($ppent/dp$) times the proportion of property, plant and equipment in total assets ($ppent/act$), plus current assets over the cost of goods sold ($act/cogs$) times the proportion of current assets in total assets (act/act); we cap it at 25 years
Abnormal earnings	The difference between the income before extraordinary items, adjusted for common stock equivalents ($ibad-1_ibad$) over the market value of equity used in calculating earnings per share ($prcc_f+cshp$)
Investment skewness	The firm-level skewness of investment, measured as the

Table C.1 part 1

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Table C.1 (continued)

Variable	Definition
Investment kurtosis (firm-level)	The firm-level kurtosis of investment, measured as the ratio of capital expenditures ($cappx$) over lagged installed capital (i_ppcap); we require at least 5 observations per firm
Return on investment	EBITDA ($ebitda$) over total assets (act)
Debt cycle length	Number of years to the first leverage spike, between subsequent leverage spikes, or after the last spike, conditional on a minimum cycle length of 3 years
Alternative capital age (1)	Capital age calculated as in Subsection 3.1, except that, when the firm disinvests, the oldest vintages are disposed of first, rather than all vintages equally
Alternative capital age (2)	Accumulated ($dpcact$) to current (dpc) depreciation expense
Alternative capital age (3)	The weighted average of capital vintages, when averaging over the past T and where more weight is put on younger vintages, following Ai et al. (2012) with $T = 7$
Alternative depreciation rate	Depreciation expense (dpc) over net plant, property and equipment ($ppent$), winsorized at 1% and 99% levels before calculating capital age
Alternative capital age (4)	Capital age calculated as in Subsection 3.1 using the depreciation rate from Compustat
Alternative depreciation rate excluding amortization	Depreciation expense (dpc) minus amortization of intangibles (am) over net plant, property and equipment ($ppent$), missing amortization values replaced with zero, winsorized at 1% and 99% levels before calculating capital age
Alternative capital age (5)	Capital age calculated as in Subsection 3.1 using the depreciation rate excluding amortization from Compustat

Table C.2
Summary statistics. The table contains the summary statistics of the variables used in the regression models of net leverage and debt maturity. The sample period is from 1975 to 2018. All variables are winsorized at 1% and 99% levels and defined in Table C.1.

	Mean	Std. dev.	Q1	Median	Q3	N
Depreciation rate (BEA)	0.085	0.029	0.068	0.081	0.098	68833
Profitability	0.142	0.073	0.091	0.134	0.183	68833
Size	5.469	1.978	4.093	5.468	6.825	68833
Market-to-book	1.424	0.763	0.693	1.103	1.620	68833

Table C.1 part 2

Firm age	19.198	17.280	6.674	14.085	25.674	66739
Asset maturity	10.154	7.198	4.390	8.416	14.634	67109
Abnormal earnings	0.006	0.188	-0.020	0.009	0.035	66938
Inv. skewness	0.992	0.867	0.391	0.918	1.515	4387
Inv. kurtosis	3.755	2.692	2.116	2.903	4.446	4387
Return on investment	0.142	0.073	0.091	0.134	0.183	68833
Alternative capital age (1)	5.753	2.931	3.564	5.336	7.506	66206
Alternative capital age (2)	5.782	3.471	3.351	5.173	7.448	68125
Alternative capital age (3)	3.350	0.758	2.907	3.361	3.796	52697
Depreciation rate (Compustat)	0.196	0.260	0.099	0.138	0.202	74039
Alternative capital age (4)	5.511	2.779	3.493	5.116	7.031	74039
Depreciation rate excl. amortization	0.170	0.176	0.096	0.132	0.185	74039
Alternative capital age (5)	5.639	2.748	3.628	5.250	7.159	74039

C.3. Summary statistics

Table C.2 contains the summary statistics of all the variables used in the paper which were not provided in Table 1.

Table C.2

6 Short summary

- The paper proposes a theory of debt cycles and maturity matching driven by finite asset life and planned replacement investment.
- Firms borrow to finance investment, then delever as capital ages to free debt capacity for future replacement.
- Debt maturity is chosen so that repayment frees up capacity when replacement investment is due.

- The model predicts that capital age is negatively related to leverage and debt maturity within firms.
- The model predicts that useful asset life is positively related to debt cycle length and debt maturity across firms.
- Using U.S. public firms from 1975 to 2018, the authors find strong support for the within-firm capital-age predictions.
- Mechanism tests show stronger effects when investment is lumpier and return on investment is lower.
- Robustness tests show that results survive alternative samples, depreciation definitions, and capital-age measures.
- Cross-sectional tests show that useful life predicts debt cycle length and average debt maturity.

7 Conclusion

7.1 Core conclusion

The paper’s central conclusion is that asset replacement needs are a first-order determinant of leverage and debt maturity. Firms do not simply choose leverage based on static trade-off variables. They manage debt capacity dynamically because capital ages and must eventually be replaced.

7.2 Economic intuition

A firm with newly installed capital has just paid a large investment cost and can use debt to finance it. But as the asset gets older, the firm knows another large investment is approaching. The firm therefore repays debt or builds cash to restore financing capacity. Debt maturity is chosen to support this process: liabilities should mature or amortize when the firm needs capacity for replacement investment.

7.3 Why this changes the interpretation of maturity matching

Traditional maturity-matching arguments often focus on matching the maturity of debt to the asset being financed. This paper adds a dynamic layer: the relevant maturity is not only about the asset’s current life, but about the firm’s position in the replacement cycle. A young asset can support longer debt; an old asset should be associated with shorter remaining debt maturity.

7.4 Why capital age and useful life are different

Capital age is a time-series state variable. It tells us how close a firm is to its next replacement investment. Useful life is a cross-sectional horizon variable. It tells us how long the replacement cycle tends to be. Separating these concepts is the paper’s key empirical insight and explains why firm fixed-effect regressions should focus on capital age rather than asset maturity.

7.5 Empirical takeaway

The evidence supports the model on several dimensions: capital age predicts lower leverage, capital age predicts shorter debt maturity, the effects are stronger when investment is lumpy and lower-return, and useful life predicts debt cycle length and maturity across firms. The consistency across these tests is more persuasive than any single coefficient.

7.6 Limits and open questions

The empirical tests are not randomized and capital age may still be correlated with unobserved time-varying factors. The model abstracts from many real-world features including rating constraints, tax dynamics, covenant renegotiation, credit lines, and market-timing motives. Future work could combine capital-age dynamics with refinancing risk and debt contract heterogeneity.

7.7 How to use the paper

The paper is useful as a framework for thinking about financing around replacement investment. It suggests that researchers studying leverage, debt maturity, investment spikes, or debt cycles should control for where firms are in the capital-age cycle. It also suggests that empirical maturity-matching tests should not treat asset maturity as a single static variable.

7.8 One-sentence takeaway

Firms match debt to assets not only because assets have lives, but because aging assets create predictable future financing needs that shape leverage and maturity today.